

Ngoc Hung Nguyen

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RESEARCH INTEREST

Computing: Cloud/edge computing, IoT, deadline-aware task scheduling
Intelligence: Applied AI/ML for networking (deep learning, deep reinforcement/reinforcement learning)
Algorithms: Greedy, graph-based algorithms, evolutionary computing.

EDUCATION

- Lund University** Period: 2025 - 2029
Toward Ph.D at the department of Electrical and Information Technology
Topic: Algorithms for XR in 6G networks
Supervisor: Professor Bjorn Landfeldt, Ph.D
Lund, Sweden
- Hanyang University** 09-2021 - 02-2024
M.Sc. in Electrical and Electronic Engineering
Thesis: Offloading under Hard Deadline in Mobile Edge Computing
Supervisor: Professor Sang-Woon Jeon, Ph.D
Ansan, Korea
- Hanoi University of Science and Technology** 09-2013 - 08-2018
Engineer degree of Chemical engineering
Thesis: Applied Numerical Methods in Chemical Engineering
Supervisor: Professor Nguyen Dang Binh Thanh, Ph.D and Professor Bui Minh Dinh, Ph.D
Hanoi, Vietnam

WORK AND ACADEMIC EXPERIENCE

- Phenikaa University**  2025–Present
Researcher and Lecturer (currently on temporary leave)
Hanoi, Vietnam
 - Teaching: Cloud Computing, Computer Architecture, and Data Processing with Python.
 - Conducting research with students in the ICNL Lab.
- VinUniversity**  03-2024 - Present
Research Assistant
Hanoi, Vietnam
 - Research on Intelligent Transportation Systems using metaheuristics and Deep Reinforcement Learning techniques.
 - Exploring the application of AI and IoT in aquaculture.
- FPT Software**  01-2024 - 12-2024
Senior AI Engineer
Hanoi, Vietnam
 - Applied quantization methods to classification and detection models, including Reset, SSD-Resnet, and DERT.
 - Developed quantized pipelines incorporating QAT, PTQ, and TVM_CPU frameworks.
 - Conducted testing and performance analysis of AI models.
- Hanyang University**  09-2021 - 02-2024
Research Assistant
Ansan, Korea
 - Research focused on Wireless Communication and Mobile Edge Computing.
 - Applied Deep Learning and Reinforcement Learning techniques to solve optimization problems.
 - Developed solutions using greedy algorithms for optimization.
- LG Electronics**  10-2018 - 07-2021
Senior Embedded Engineer
Hanoi, Vietnam
 - Developed Wi-Fi middleware as part of the Wi-Fi core framework.
 - Created SPI middleware for the SPI-Fi core framework.
 - Performed unit testing using Google Test framework.
 - Developed a tool for generating unit test code (C/C++ using Qt Creator on Windows OS).
 - Researched methods for evaluating camera sharpness based on different regions, leading to the development of test software for camera performance.
 - Worked on autofocus and low contrast brightness improvement features.
 - Developed various software tools for testing and performance analysis.

HONORS AND AWARDS

- Third-award - Student Research Prize** Hanoi
Hanoi University of Science and Technology 2017-2018

SKILLS

- **Programming Languages:** Python, C++/C, Cuda-C++, MATLAB
- **Data Science & Machine Learning:** Pytorch, TensorFlow, Pandas, Numpy
- **Technique:** Mathematical Modeling, Numerical Methods, Optimization, Applied AI/ML (e.g., RL/DRL)
- **Languages:** Vietnamese (mother tongue), English (fluent)

ACADEMIC SERVICES

- **Journal reviewer:** IEEE Internet of Things, IEEE Communications Letters, IEEE Transactions on Intelligent Transportation Systems, and Computer Communications

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

Research profiles:

- Google Scholar: <https://scholar.google.com/citations?user=uOr3eosAAAAJ&hl=vi&oi=ao>
- ORCID: <https://orcid.org/0009-0007-7363-5014>
- ResearchGate: <https://www.researchgate.net/profile/Nguyen-Hung-120>

- [J.1] **Ngoc Hung Nguyen**, Nguyen Van Thieu, Senura H. Wanasekara, Van-Dinh Nguyen, Quang-Trung Luu, Nguyen Cong Luong, and Anh Tuan Nguyen, "**Oranits: Autonomous Control and Task Allocation in ITS Integrating MEC and Open RAN using Metaheuristic and Deep Reinforcement Learning**," *IEEE Transactions on Intelligent Transportation Systems*, 2026, DOI: 10.1109/TITS.2026.3705751.
- [s.1] **Ngoc Hung Nguyen**, Van Thieu Nguyen, and Bjorn Landfeldt. "MoRo: Mobility-Aware Robot Mission Orchestration under Computation-Induced Motion Degradation." Submitted to Proceedings of the IEEE International Conference on Computer Communications (INFOCOM).
- [s.2] **Ngoc Hung Nguyen** and Bjorn Landfeldt. "Learning-Based Collaborative MEC for LLM Inference with Soft-Deadline Awareness via Transformer-Enhanced PPO." Submitted to Proceedings of the IEEE Global Communications Conference (GLOBECOM).
- [c.1] **Ngoc Hung Nguyen**, Van Thieu Nguyen, Quang-Trung Luu, Phi Son Vo, and Van-Dinh Nguyen. "**A Metaheuristic Approach for Mission Assignment and Task Offloading in Open RAN-Enabled Intelligent Transport Systems**," in Proceedings of the IEEE Global Communications Conference (GLOBECOM), Taipei, Taiwan. IEEE, 2025.
- [J.2] Tran Cong Dao, Nguyen Cong Luong, **Ngoc Hung Nguyen**, Xingwang Li, Dusit Niyato, and Dong In Kim, "**Multi-Hop Routing for IoT-Based Digital Twin: Novel Metaheuristic Approaches**," in *IEEE Internet of Things Journal*, vol. 12, no. 15, pp. 30493-30506, 1 Aug.1, 2025
- [J.3] Hiep, Dao Quang, Nguyen Cong Luong, Shimin Gong, Xingwang Li, **Ngoc Hung Nguyen**, Dusit Niyato, and Dong In Kim. "**Dynamic Multi-layer Aerial System for Latent Diffusion-based Generative AI Inference at the Edge**". *IEEE Transactions on Communications*.
- [J.4] **Ngoc Hung Nguyen**, Van-Dinh Nguyen, Anh Tuan Nguyen, Nguyen Van Thieu, Hoang Nam Nguyen, and Symeon Chatzinotas, "**Deadline-Aware Joint Task Scheduling and Offloading in Mobile-Edge Computing Systems**," *IEEE Internet of Things Journal*, vol. 11, no. 23, pp. 33282–33295, Oct. 2024, DOI: 10.1109/JIOT.2024.3425854.
- [J.5] Nguyen Van Thieu, **Ngoc Hung Nguyen**, and Ali Asghar Heidari, "**Feature Selection Using Metaheuristics Made Easy: Open Source MAFESE Library in Python**," *Future Generation Computer Systems*, vol. 152, pp. 340–358, Jun. 2024, DOI: 10.1016/j.future.2024.06.006.
- [J.6] Nguyen Van Thieu, **Ngoc Hung Nguyen**, Mohsen Sherif, Ahmed El-Shafie, and Ali Najah Ahmed, "**Integrated Metaheuristic Algorithms with Extreme Learning Machine Models for River Streamflow Prediction**," *Scientific Reports*, vol. 14, no. 8, pp. 340–358, Aug. 2024, DOI: 10.1038/s41598-024-63908-w.
- [P.1] **Ngoc Hung Nguyen**, Sang-Woon Jeon, and Kangyu Gao, "**Job Scheduling with Deadline Constraints**," Patent No. 10-2023-0035648, Registration Date: Mar. 20, 2023.
- [C.2] Senura Hansaja Wanasekara, Han Huy Dung, **Ngoc Hung Nguyen**, and Van-Dinh Nguyen, "**Lossy Compression of Multi-channel EEG and PPG Signals Based on Golomb-Rice Coding with Parameter Estimation**," in *Proc. International Conference on Advanced Technologies for Communications (ATC)*, 2024.

REFERENCES

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Research Leader

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Ph.D Advisor